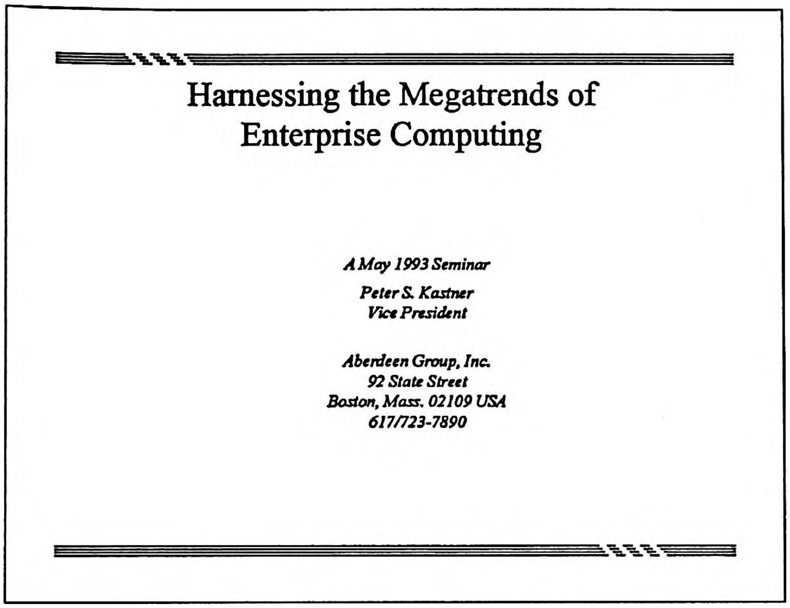
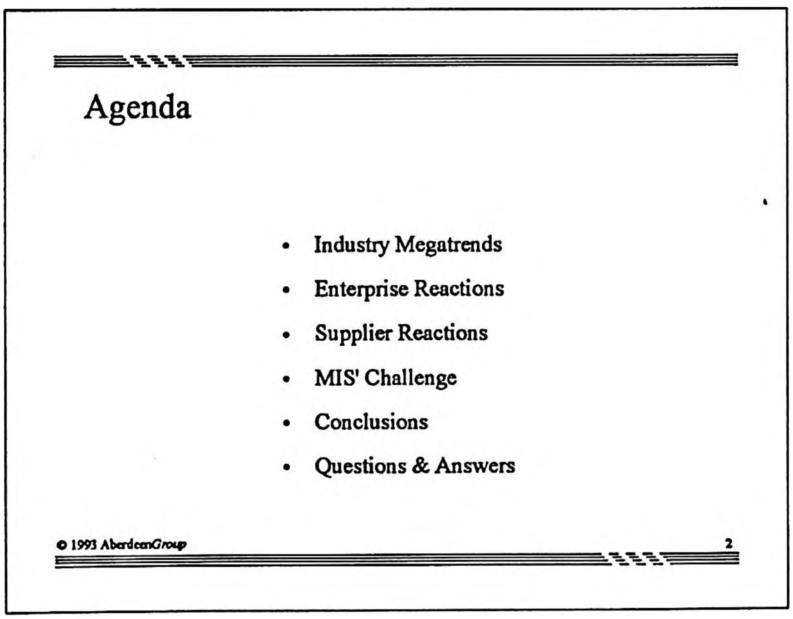
PHELPSBW

Harnessing the Megatrends of Enterprise Computing



Agenda



Enterprise Infrastructure

Enterprise Infrastructure

- Up to 50% of capital expenditure is now for IT
- Information is part of products and services
- All employees are expected to be computer-comfortable
- Online data for real-time decisions is the standard -- not the goal

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Users of Information Technology

Users of Information Technology

- The majority of end-users and operational managers are computer literate
- Department managers will determine how their enterprise applications work
- Customers and suppliers expect safe electronic interchange
- Users cannot do the jobs without computers and communications

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Technology Advances

Technology Advances

- Microprocessor speeds continue rapid performance increases
- Midrange I/O channel speeds surpass mainframes
- Desktop software matures while OS options multiply
- Object technology becomes a requirement 3-5 years out
- Higher speed, standardized networking

1992 Large Broker PC Purchases

MIPS	CPUs	Cost	$/MIPs
95,000	5,100	$22.8M	250

New Application Budgets

New Application Budgets

- For large, complex new application installations:
- 50% hardware and applications software
- 50% services
  - Planning
  - Project management
  - Network implementation
  - Application development
  - Systems integration
  - Data conversion
  - Training and education

Suppliers

Suppliers

- Less profitable
- Putting more support responsibility on customers
- Large hardware suppliers are obtaining up to 50% of their revenues from services
- Is Computer Associates the model for legacy product support?
- Desktop, software, and communications supplier confusion continues

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MIS Role in the 1990s

MIS Role in the 1990s

- MIS is a corporate capability to win — or lose
- MIS is primary force to lower overhead expenses
- Tools be implemented (and re-implemented) that make end-users more effective
- Communications becomes more important than programming
- MIS provides more support to workgroups

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Enterprise Financial Issues

Enterprise Financial Issues

- BigCompany Income Statement
  - 100% Revenues
  - 55% Cost of goods sold
  - 35% Selling, general and administrative
  - MIS is 1%-5% of Revenues
  - 10% Profit before taxes

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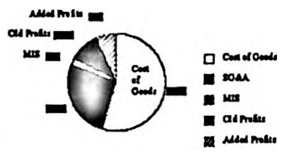
Leveraging Information System The Key to Profits in the 1990s

Leveraging Information Systems
The Key to Profits in the 1990s

- If MIS lowers SG&A by 20% ⟶ PBT increases 70%
- If MIS lowers COGS by 5% ⟶ PBT increases 30%
- If MIS cuts MIS costs by 50% ⟶ PBT increases by 15%



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Enterprise Reactions

Enterprise Reactions

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Enterprise Reactions

Enterprise Reactions

- Desire to lower overhead costs by 20% to improve profits by ~50% using new technologies
- Re-engineer front-office processes and applications to match/establish strategy
- Do not invest in mainframes for the long-term
- Find alternatives to perennial 18-36 month application backlog

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Enterprise Reactions

Enterprise Reactions

- Cut MIS budgets without lowering service requirements
- Empowerment of user workgroups
- Senior management begins investigating MIS alternative strategies
- Success by the skillful — frustration for others

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Challenges to MIS

Challenges to MIS

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Evolution or Revolution?

Evolution or Revolution?

- Mainframe applications are old, undocumented, expensive to maintain, slow to change
- New generation wants to leave its mark
- Legacy systems contain the data and procedures that are running the business
- Procedures need to be replaced

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Barriers to Internal MIS Transition

Barriers to Internal MIS Transition

- Lack of confidence in/experience with today's leading-edge systems software tools
- Too few trained technical staff resources
- Not enough ISV applications — barely moved to RDBMS much less GUI-based client-server
- Total initial implementation costs seem too high
- Requirements for self-reliance are very high

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Barriers to MIS Organization Transforming Role

Barriers to MIS Organization Transforming Role

- Lack of an enterprise strategy — or too many strategies
  - MIS forced to establish strategy or move slowly
  - MIS not prepared to set strategy
- Many mainframers find the transition difficult or impossible
- Requires new supplier-partner relationships

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MIS Challenges

MIS Challenges

- Develop the next generation enterprise-wide Information Architecture taking advantage of both open systems and client-server technologies
- Train internal development staffs on today's best open systems, client-server computing products
- Budget out through the next five years for capital and training costs versus benefits
- Begin implementing today!

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And the Challenge is Well Recognized

And the Challenge is Well Recognized

"Simply put, the flip side of the new super-low-cost computing units being delivered by the new computing industry is a huge integration task -- but also a tremendous business opportunity. The smart companies are laying the foundation now for doing this kind of integration work."

Andy Grove, CEO of Intel Corporation, *Wall Street Journal*, 18 January 1993

Why Mainframe Alternatives Now?

Why Mainframe Alternatives Now?

- Need to lower overhead costs and improve profits
- Front-office processes and applications need to match/establish strategy
- New mainframes are not good long-term investment
- If MIS is the competitive advantage of the 1990s, mainframes do not provide a differentiator

Evaluating Alternatives Now

Evaluating Alternatives Now

- Cost to operate and upgrade
- Application backlog
- Global enterprise-wide information systems
- User empowerment
- Midrange and desktop technology has surpassed mainframes
- MIS ability to execute

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Mainframe Alternatives

Mainframe Alternatives

- Consolidate data centers
- Cheap MIPS — used or non-IBM
- Outsource — particularly static applications
- Freeze/lower MIS budgets
- Downsize applications with replicated systems
- Non-MIS divisional/departmental installations

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Downsizing

Downsizing

- Connotations negative but realistic
- Replicated, distributed systems
- Lower costs and/or increase enterprise functionality
- MIS gains control of departmental systems and PCs (whether it wants them or not)
- Requires new MIS management mode — business strategy, fast moving, flexible

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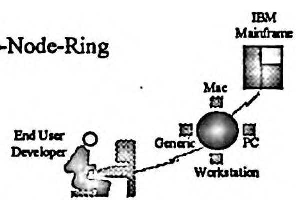
Enterprise Information Architecture

Enterprise Information Architecture

Spoke-Node-Ring



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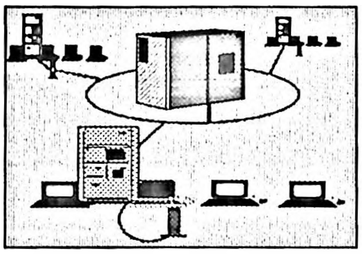
Enterprise Topology

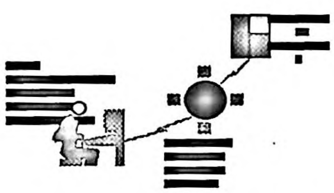
Enterprise Topology



- Three-tier plus is state-of-the-art

Enterprise Topology Basics

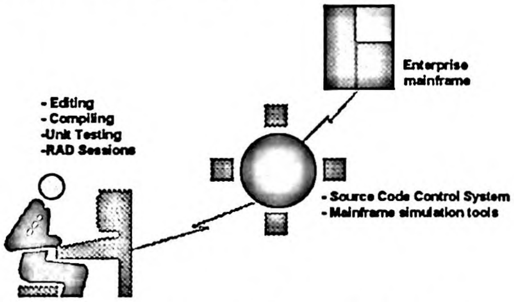
Enterprise Topology Basics



- The next generation of departmental open, client-server systems

Development Topology Basics

Development Topology Basics



Up to 20% of mainframe workloads can be eliminated by moving development to PC LANs.

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It's Not Just Client-Server !

It's Not Just Client-Server !

- GUI and integated tool sets
- Automate automation
- Leverage off-the-shelf applications
- Let the users help

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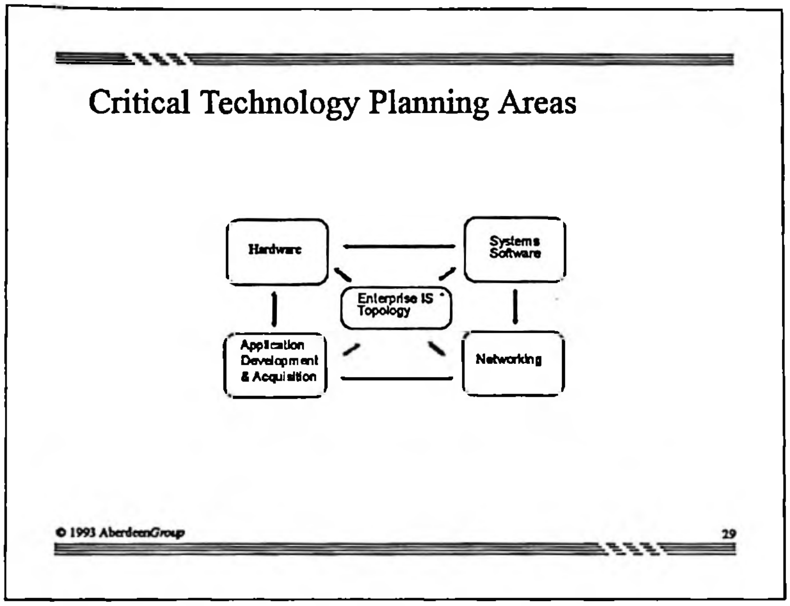
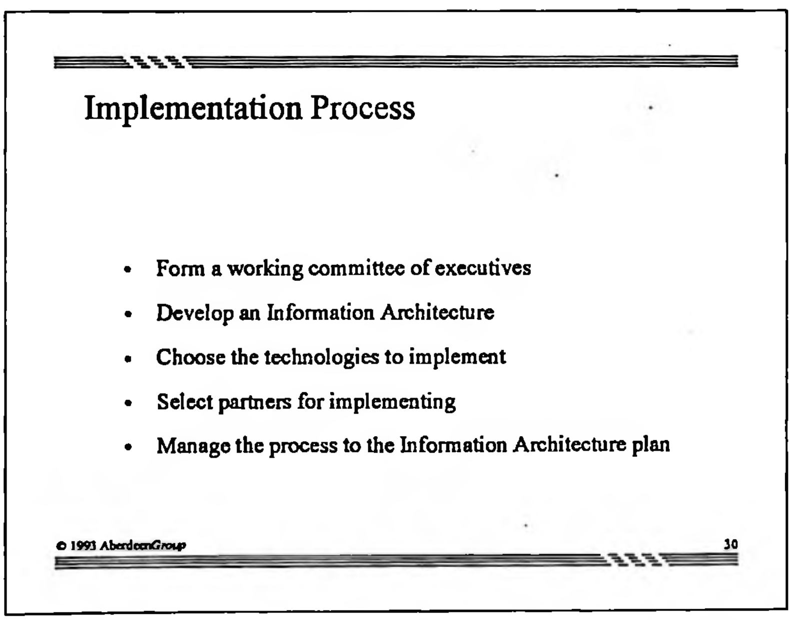
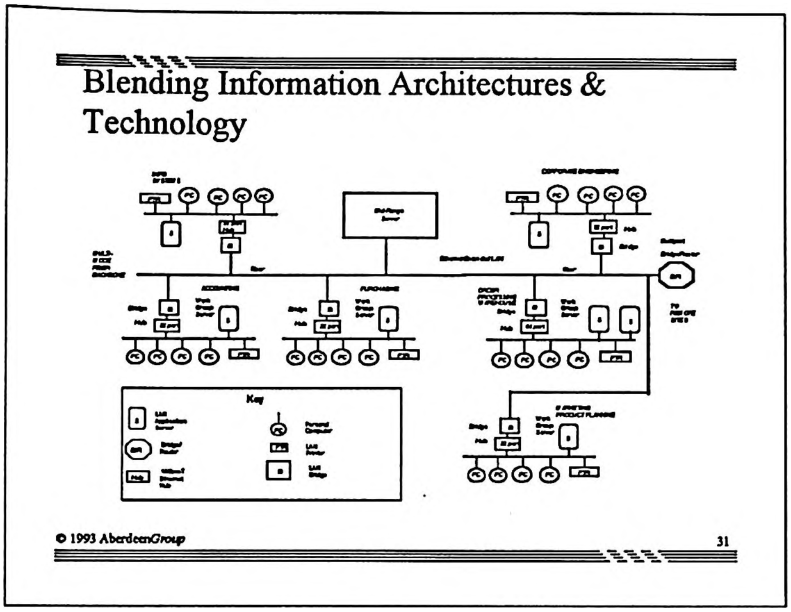
Critical Technology Planning Areas



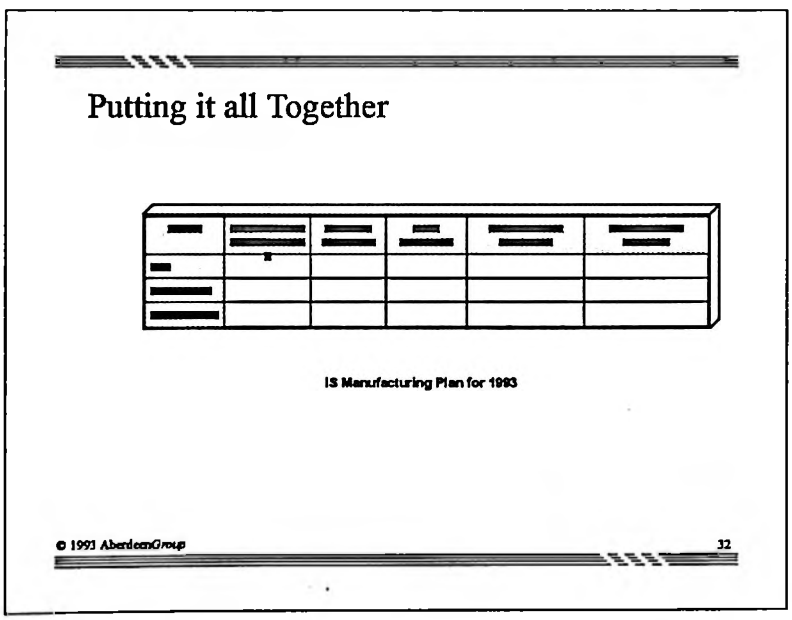
Implementation Process



Blending Information Architectures & Technology



Putting it all Together



Suppliers

Suppliers

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Hardware

Hardware

- IBM and Plug-Compatibles
- Unisys
- Hewlett-Packard
- NCR
- Digital Equipment
- Sun Microsystems
- Compaq and Dell

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Enterprise Software

Enterprise Software

- Computer Associates
- Legent, Candle, etc.
- Dun & Bradstreet Software
- CASE suppliers

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Operating Systems

Operating Systems

- Will there ever be one Unix?
- Windows 3.1
- Windows NT
- OS/2
- NetWare

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Databases

Databases

- IBM
- Oracle
- Sybase
- Ingres
- Informix
- Software AG

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Client-Server Development

Client-Server Development

- Powersoft, Gupta and Blyth
- RDBMS suppliers
- DOS suppliers grow up: Borland, Lotus, Software Publishers
- Micro Focus
- Microsoft

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Wrap-Up

Wrap-Up

- The skills required for change are the skills of management and business
- Information Architecture is the integrated automation of the strategies and policies of the enterprise
- Downsizing technology is embodied in standardized tools that typical MIS staff can use
- Adopting new technologies is simply a business process -- and all the processes seem difficult the first time
- Skilled partners can shorten implementation time and increase quality

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Wrap-Up

Wrap-Up

- Technology is not a barrier to change
- There is no single application that cannot be effectively run on midrange systems
- RDBMS is the multi-user, technology transport for the future
- Mainframe consolidation and outsourcing are temporary alternatives to downsizing
- Cutting MIS budgets alone has a poor ROI

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